

Deep SQL Analysis

Project

Deriving Actionable Insights Through Advanced SQL
Queries

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Project Overview

A comprehensive and structured approach to analyzing business datasets using SQL. This repository provides specialized scripts designed for performing in-depth data analysis across multiple critical business domains.

Focusing on customer behavior, product performance, overall sales trends, geographical metrics, and advanced statistical methodologies.

| Customer Insights



General Analysis

Execution of

Customer_Analysis.sql

to establish a baseline understanding of overall customer data, demographics, and high-level engagement metrics.



Purchase Patterns

Tracking detailed customer buying behaviors using

Customer_Purchase.sql

to identify frequency, average order value, and purchasing habits.



Cohort Tracking

Utilizing

Customer_Cohort_Analysis.sql

for analyzing customer retention rates and behavioral shifts over specific time periods through cohorts.

| Product & Category Analysis



Product Level

Detailed evaluation of individual product performance using [Product_Analysis.sql](#).

This script dives into specific SKUs to determine top sellers, underperforming items, and overall inventory health based on historical transaction data.



Category Level

Broader insights into sales and performance aggregation via

[Product_Category_Analysis.sql](#).

Understanding which departments or product families drive the most value, helping to guide high-level merchandising and procurement strategies.

| Performance & Trends



Overall Health

Overall_Performance.sql

provides a high-level executive summary of absolute business health, combining key revenue and operational metrics.



Sales Trends

Sales_Trend_Analysis.sql

identifies seasonal patterns, growth trajectories, and cyclic anomalies over specific, defined time periods.



Geographical Data

Country_Analysis.sql

breaks down business metrics by region, allowing for localized strategy adjustments and market penetration analysis.

| Statistical Analysis

80/2

The Pareto Principle

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Identifying Key Drivers

The repository features specialized scripts to apply statistical methods directly within SQL.

`Pareto_80-20_Analysis.sql` provides a general implementation of the principle across various datasets.

`Product_Pareto_Analysis.sql` specifically identifies the vital 20% of products that reliably generate 80% of the total revenue.



1. Clone Repository Clone the `alsoub-labs` repository to your local development environment or server.



2. Database Access Ensure access and connection to a SQL-compatible relational database (e.g., PostgreSQL, MySQL, or SQL Server).



3. Data Import Safely import your raw business data and schemas into the target database.



4. Execute Scripts Run the desired `.sql` script files against your database to automatically generate the analysis output.

| Project Information



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Lead Contributor

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